Optimal Packing of Fifteen Equal Disks in a Circle: An Exact Computer-Assisted Certificate

October 11, 2026

Abstract

We study the minimum radius of a circle containing fifteen pairwise interior-disjoint unit disks. We give an exact construction of radius

$$R_0 = 1 + \sqrt{1 + \big(2 + \cot(\pi/5)\big)^2} = 4.521356964706164\ldots$$

and a candidate computer-assisted certificate for the matching lower bound. A radial threshold reduces every putative packing with radius at most R_0 to four possible inner/outer counts. For each of the 760 dihedral classes of their cyclic binary words, an integer difference-constraint test either excludes the class or passes it to exact rational interval refinement. Refinement leaves one cyclic type, $(I, O, O)^5$, and confines its five inner-center radii to $[1.68, 1.72]$. An explicit local angular inequality then forces the candidate radius vector and excludes every smaller container. The certificate uses integer and rational arithmetic, including an explicit Taylor remainder bound for the trigonometric table. The repository contains a serialized subdivision certificate, a separate rational checker, and a one-command replay.

Verification status. The finite computations and the separate certificate checker were successfully replayed with the accompanying source bundle. A subsequent Lean development checks the finite certificate replays and formalizes the local angular obstruction, together with parts of the geometric reduction. It does not yet prove that every feasible packing reaches the final local box, or construct the feasible candidate packing as a Lean object. This manuscript presents a candidate computer-assisted proof; a complete Lean proof of its main theorem is still outstanding. The scope and trust dependencies of the partial formalization are recorded in section B.

1 Introduction

Packing congruent disks in a circle is equivalent to maximizing the minimum distance among a fixed number of points in a disk. Graham proved the small cases through seven points, and Pirl treated the circle-packing cases through ten. Later computational work by Graham, Lubachevsky, Nurmela, and Ostergård tabulated best-known configurations and described the configurations for 12 through 20 disks as conjecturally optimal; their 15-disk entry is the familiar five-inner, ten-outer arrangement [1, 2, 3]. Ekanayake and LaFountain subsequently proved the 14-disk case and reported that the proved cases then known were $N \leq 14$ and $N = 19$ [4]. The 15-disk case is therefore a natural next problem.

Proof overview. The main difficulty is to exclude nonsymmetric packings without assuming an optimizer's contact graph. We first replace each outer center by a wall center at the same polar angle, then convert pairwise separation into angular difference constraints. An exact finite certificate checks all 760 cyclic order classes and leaves one type, $(I, O, O)^5$, with five inner radii in a small interval. In that interval, the direct inner-pair constraint and the route through the two intervening outer centers respond in opposite ways to radial perturbations. Their maximum creates a strict angular penalty away from the symmetric candidate. The finite stage rejects a radial box only when a necessary condition fails; a surviving box is not treated as evidence that a packing exists.

2 Problem and main result

Let $p_1, \dots, p_{15} \in \mathbb{R}^2$ be the centers of fifteen unit disks. A circle of radius R centered at the origin contains the disks without interior overlap exactly when

$$\|p_i\| \le R - 1 \quad (1 \le i \le 15), \qquad \|p_i - p_j\| \ge 2 \quad (i \ne j). \tag{1}$$

Write R_{15}^* for the infimum of feasible R.

Theorem 2.1. *The minimum enclosing radius for fifteen pairwise interior-disjoint unit disks is*

$$R_{15}^* = R_0 = 1 + \sqrt{1 + \big(2 + \cot(\pi/5)\big)^2}.$$

This is the conclusion claimed by the candidate argument below. It is not yet the conclusion of an unconditional Lean theorem; in particular, the packing-level implication in proposition 6.1 remains to be connected to the checked finite certificates.

The upper bound comes from the familiar D_5-symmetric arrangement, invariant under rotations by $2\pi/5$ and reflections, with five inner disks and ten outer disks; it is listed among the best-known configurations [3, 5]. The lower-bound strategy is independent of the tight-partition reduction used for fourteen disks [4].

3 An exact packing of radius R_0

Set

$$\varphi = \frac{\pi}{5}, \qquad a_0 = \csc\varphi, \qquad b_0 = \sqrt{1 + \big(2 + \cot\varphi\big)^2}, \qquad \alpha = \varphi - \arcsin(1/b_0), \qquad R_0 = 1 + b_0.$$

For $j = 0, \dots, 4$, take centers

$$I_j = (a_0, 2j\varphi), \qquad O_j^- = (b_0, 2j\varphi - \alpha), \qquad O_j^+ = (b_0, 2j\varphi + \alpha)$$

in polar coordinates. The five inner centers form a regular pentagon, with two outer centers between each pair of consecutive inner-center directions [5].

Adjacent inner centers have distance $2a_0 \sin\varphi = 2$. The smaller of the two successive angular gaps between outer centers is $2(\varphi - \alpha)$, and

$$2b_0 \sin(\varphi - \alpha) = 2.$$

The other outer gap is larger because $\alpha > \varphi/2$. For an inner center and either of its two nearest outer centers, the cosine rule gives

$$a_0^2 + b_0^2 - 2a_0 b_0 \cos\alpha = 4.$$

These are the closest pairs within each type; all other same-type or mixed pairs have a larger angular separation and hence a larger center distance. Every outer disk is tangent to the container boundary, and $a_0 < b_0$. Thus the arrangement is feasible in radius R_0, proving $R_{15}^* \le R_0$; figure 1 shows the resulting placement.

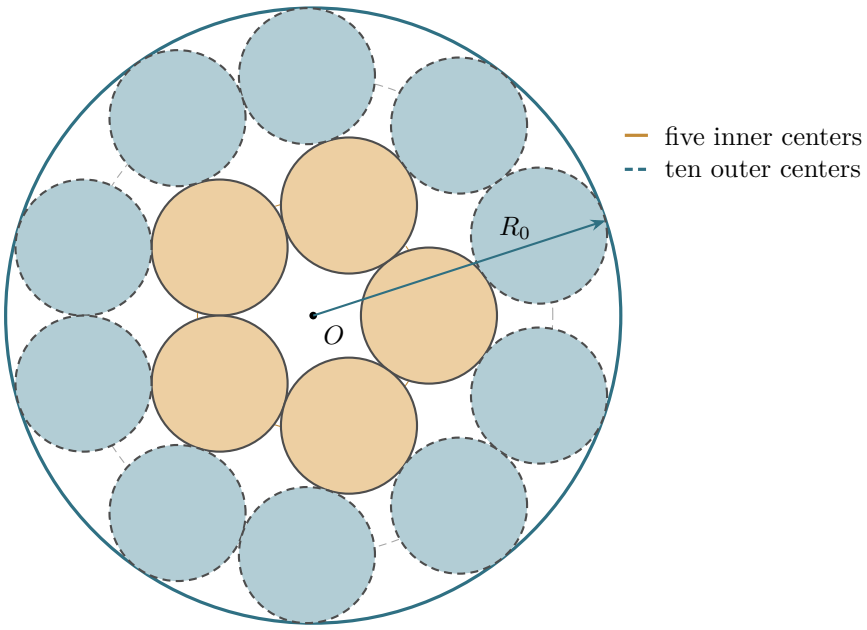


Figure 1: The exact fivefold-symmetric construction. Inner disks have solid outlines and outer disks dashed outlines, so the two types remain distinguishable in grayscale. The dashed concentric circles mark center radii. This verifies the upper bound only and makes no symmetry assumption about a minimizer.

4 Radial reduction

For a center at distance s from the origin, call it inner if $s < L$ and outer if $s \ge L$, where

$$L = b_0 - \frac{4}{b_0}. \tag{2}$$

The decimal enclosures used by the verifier are rational outward bounds; in particular $2.385431 < L < 2.385432$ and $b_0 < 3.522$.

For $x, y > 0$ define

$$c(x,y) = \frac{x^2 + y^2 - 4}{2xy}, \qquad \Phi(x,y) = \arccos c(x,y)$$

whenever $|x - y| < 2 \le x + y$. If the two center radii are x, y and their smaller angular separation is $\Delta \in [0, \pi]$, non-overlap requires $\Delta \ge \Phi(x,y)$.

These are radial bookkeeping labels; they do not assume that an optimizer has five centers of one kind and ten of the other. The threshold is chosen so the mixed corner (L, b_0) and the wall corner (b_0, b_0) require the same angle:

$$c(L, b_0) = c(b_0, b_0).$$

This identity is the key to the outer–outer comparison in the wall-replacement lemma below.

Lemma 4.1 (Four-corner maximum)**.** *On every positive rectangle $[u_1, u_2] \times [v_1, v_2]$, the maximum of $c(x, y)$ is attained at a corner.*

Proof. For fixed y,

$$\frac{\partial c}{\partial x} = \frac{x^2 - y^2 + 4}{2x^2 y}.$$

Its numerator is strictly increasing in x, so the derivative can change sign only from negative to positive. Thus an interior critical point in the x direction is a minimum, and a maximum occurs at an endpoint. Apply the same argument in y. □

Lemma 4.2 (Replacing outer centers by wall centers)**.** *Suppose a configuration satisfies (1) for some $R \le R_0$. Simultaneously replace every outer center of radius r by a center of radius b_0 at the same polar angle. The resulting configuration satisfies all pairwise non-overlap constraints and is feasible in a container of radius R_0. This is a statement about the final configuration; it does not assert that a continuous motion between the two configurations stays feasible.*

Proof. For an inner–outer pair, let $x < L \le y \le b_0$ be the inner and outer radii before replacement. Its squared distance at fixed angular difference is increasing in the outer radius, since

$$\frac{d}{dy}\big(x^2 + y^2 - 2xy\cos\Delta\big) = 2(y - x\cos\Delta) > 0,$$

so the pair remains separated after the outer radius is replaced by b_0. For an outer–outer pair, lemma 4.1 and $L = b_0 - 4/b_0$ show that

$$\max_{L \le x, y \le b_0} c(x, y) = c(b_0, b_0),$$

with equality also at (L, b_0) and (b_0, L). Feasibility before replacement gives $\cos\Delta \le c(x, y) \le c(b_0, b_0)$, so this pair is also separated in the resulting configuration. □

Let $\theta_0 = 2\arcsin(1/b_0)$. Any two outer centers, after the preceding move, require angular separation at least θ_0. Since $11\theta_0 > 2\pi$, there are at most ten outer centers. Nine inner centers would be nine unit disks inside radius $L + 1 < 3.4$, whereas the exact nine-disk minimum radius is $1 + \sqrt{2(2 + \sqrt{2})} > 3.6$ [2, 5]; hence there are at most eight inner centers. Since there are fifteen disks in all,

$$(|\mathcal{I}|, |\mathcal{O}|) \in \{(5, 10), (6, 9), (7, 8), (8, 7)\}. \tag{3}$$

The threshold and the surviving count ranges are sketched in figure 2.

Write $s_{(1)} \le \cdots \le s_{(15)}$ for the fifteen center radii in nondecreasing order. The standard small-packing bounds $s_{(5)} \ge a_0 > 5/3$ and $s_{(6)} \ge 2$ [1] are also used in the radial interval enumeration. They follow by applying the known exact five- and six-disk packing radii to the five and six innermost centers. Consequently, among k inner centers at least $k - 4$ have radius at least $5/3$, and at least $k - 5$ have radius at least 2.

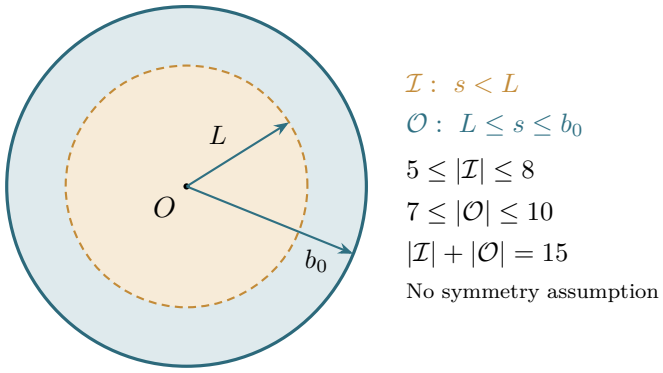


Figure 2: The radial threshold separates inner centers from the outer annulus. The count bounds leave only the four populations in equation (3); this is not a packing diagram.

5 Angular difference constraints

Order the centers by nondecreasing polar angle and lift their angles to $\theta_1 \le \cdots \le \theta_{15} < \theta_1 + 2\pi$. For a pair whose radii lie in positive intervals X_i, X_j, let

$$C_{ij} = \max\{c(x, y) : x \in X_i, \ y \in X_j\}.$$

The maximum is obtained at the four interval corners by lemma 4.1. The exact zero-radius case and positive intervals whose closure touches zero are handled separately; the endpoint rules and their bin ownership are proved in section 6.1.

If the interval box gives $x+y < 2$ for every choice of radii in a pair, that pair cannot be separated, so the box is discarded. For every pair not discarded by this test, $C_{ij} \ge -1$. If $-1 \le C_{ij} < 1$, put $\gamma_{ij} = \arccos C_{ij}$. Every feasible angle assignment must satisfy

$$\gamma_{ij} \le \theta_j - \theta_i \le 2\pi - \gamma_{ij} \qquad (i < j). \tag{4}$$

A box for which $|x - y| \ge 2$ throughout imposes no angular restriction for that pair. These rules are relaxed outwards, so every actual packing in a radius box satisfies the resulting difference constraints. The geometric conversion is illustrated in figure 3.

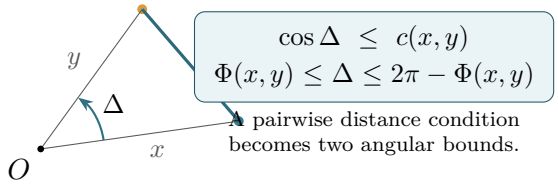


Figure 3: The law of cosines converts center separation into a lower bound on the smaller angular gap. The complementary upper bound in equation (4) is also needed for the difference-constraint graph.

Each inequality $\theta_v - \theta_u \le w$ is a directed edge $u \to v$ with weight w. Such a system is feasible only if the directed graph has no negative cycle. The verifier replaces each required angle by a certified rational lower bound $q/2800$ and uses $44/7$ as an upper bound for 2π. Thus a lower-bound edge has weight $-q$, while its complementary upper-bound edge has weight $17600 - q$. A negative cycle in these integer weights excludes the entire radial box.

For example, suppose three angle constraints require

$$\theta_2 - \theta_1 \geq \lambda, \qquad \theta_3 - \theta_2 \geq \mu, \qquad \theta_3 - \theta_1 \leq \nu.$$

The first two imply $\theta_3 - \theta_1 \geq \lambda + \mu$, so no angles can satisfy all three if $\lambda + \mu > \nu$. In the difference graph, these inequalities give a directed cycle of total weight $\nu - \mu - \lambda < 0$.

Certified trigonometric bounds. For $x \geq 0$, let $P_{18}(x) = \sum_{j=0}^{9} (-1)^j x^{2j}/(2j)!$. Taylor's theorem and $|\cos^{(19)}(t)| \leq 1$ give the rational lower bound

$$\cos x \geq P_{18}(x) - \frac{x^{19}}{19!}. \tag{5}$$

The explicit remainder subtraction is essential: the even Taylor polynomial alone is not used as a lower bound. For an interval pair with corner maximum C_{ij}, the integer q is chosen so that the right side of (5) at $x = q/2800$ is at least C_{ij}. Since cosine decreases on $[0, \pi]$, this certifies $q/2800 \leq \gamma_{ij}$. The integer range is certified by exact Machin-series bounds: $8790/2800 < \pi < 22/7$, so every tested angle is below π and $2\pi < 44/7$. All comparisons are performed with exact fractions.

6 Exhaustive finite certificate

The cyclic inner/outer word is defined up to rotation and reflection, because the disks are congruent and the container is rotationally and reflectively symmetric. Burnside's lemma gives 111, 185, 232, and 232 classes for the four inner counts in equation (3), for a total of 760. The word records only the order of the two radial labels around the circle; it does not impose geometric symmetry, as illustrated in figure 4.

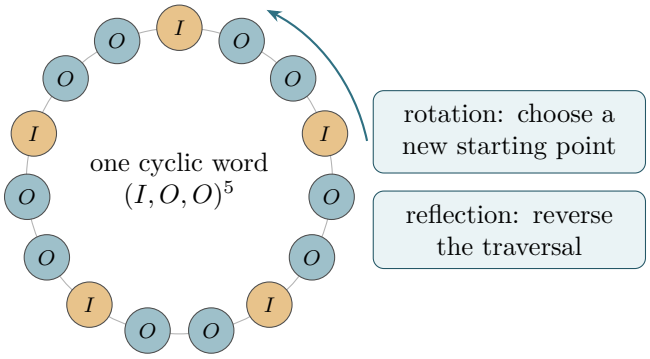


Figure 4: A binary bracelet records only the cyclic inner/outer order. Rotating or reversing the word changes its representation, not the underlying case; it does not make a geometric packing symmetric.

Proposition 6.1 (Finite reduction)**.** *After replacing outer centers by wall centers, every feasible packing of radius at most R_0 has cyclic type $(I, O, O)^5$, and its five inner-center radii lie in*

$$[42/25, 43/25]^5 = [1.68, 1.72]^5.$$

This is the conclusion of the finite certificate. A terminal box contained in this cube is passed to the analytic argument; it is not asserted to contain a feasible packing.

The finite certificate for proposition 6.1 has three stages. First, exact rational arithmetic checks a fixed four-type angular table, and an integer negative-cycle test examines all 760 words. Second, the surviving words are refined over eight radial bins and then twelve radial bins, using the rank bounds in section 4. Third, exact rational subdivision excludes each remaining radial box by a necessary condition, except that the canonical type is retained when a box lies inside the local cube. The finite computation ends there; the local analytic lemma in the next section rules out every point of that cube except the candidate radius vector.

Here a box is called closed only when the entire represented radial region is excluded by a stated necessary condition. A local leaf records a box passed to the analytic proof and makes no existence claim. The flow in figure 5 summarizes the three finite stages and the subsequent local argument.

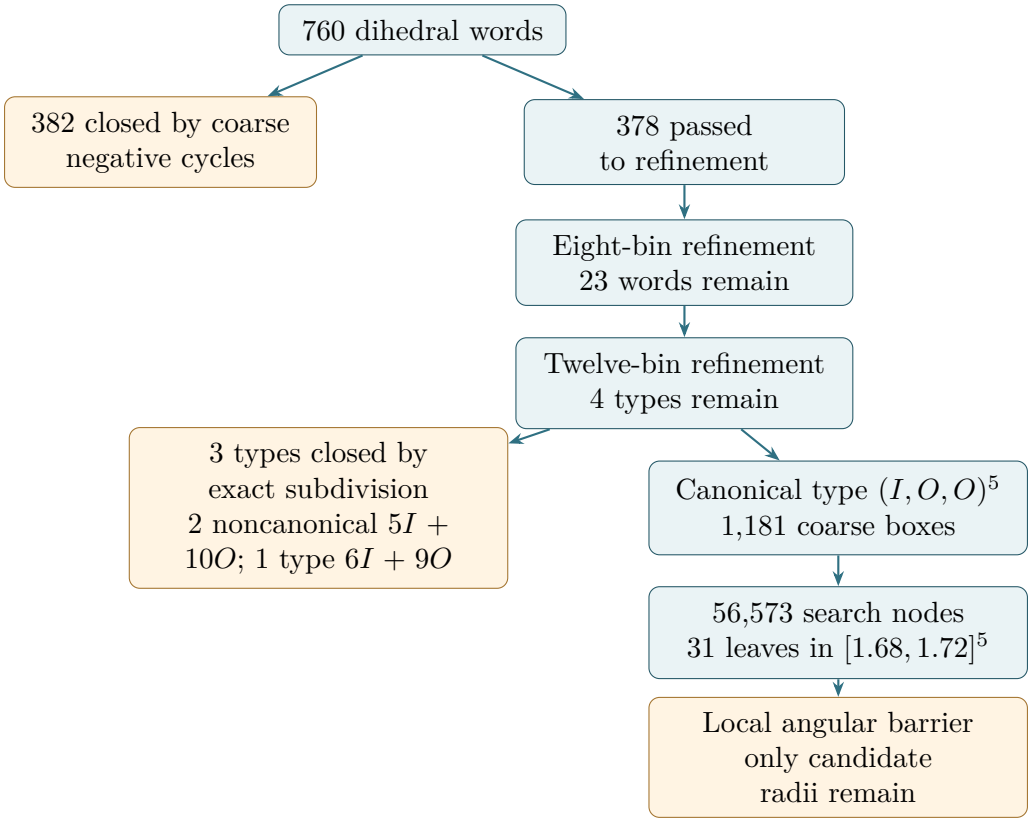


Figure 5: The first three stages form the finite reduction and test all 760 dihedral classes. The final analytic step handles the boxes passed to local rigidity.

The exact rational subdivision closes the two noncanonical $5I+10O$ words from 47 and 38 coarse radial boxes, respectively. Their branch-and-bound trees contain 1,641 and 646 nodes. The $6I+9O$ word has three coarse boxes. Splitting its unique first-bin radius into $0 < r < 1/4$, $1/4 \le r < 1/2$, and the exact origin gives nine leaves: six pair-sum contradictions and three negative cycles.

For the canonical word $(I, O, O)^5$, 1,181 coarse radial boxes generate 56,573 search nodes: 28,846 negative-cycle leaves, no pair-sum leaves, and 31 leaves passed to the local argument. The local proof below excludes every point in those boxes except the candidate radius vector, without assuming that the five radii or the packing are symmetric.

The endpoint ownership convention, origin case, and closed-hull bounds are recorded in section 6.1. The serialized data and independent certificate checks are described in the reproducibility appendix.

6.1 Endpoint ownership and the origin

The finite enumerators assign each radius to a half-open owner bin: every rational cutpoint belongs to the bin on its right. The first owner bin is split into the singleton $\{0\}$ and the positive interval $0 < r < 1/2$. The remaining owner bins begin at 1/2, 1, 3/2, 8/5, 5/3, and 2 (with the stated further subdivisions in the twelve-bin stage); the last inner owner is truncated at L. The arrays used for interval bounds are closed rational hulls of these owner bins. Consequently, the outward bounds may include shared endpoints, but the discrete enumeration neither loses an endpoint nor relies on assigning it to two different cases. At an exact bisection, the two closed child hulls cover the parent, including their common midpoint.

This convention justifies the early count tests. The labels counted by 'small' have actual radii strictly below 1, so 'small>1' gives two centers with radial sum strictly less than 2. A first-bin center and a center in $[1, 3/2)$ also have sum strictly less than 2. The labels counted by 'high' and 'high2' have lower endpoints 5/3 and 2, respectively, so the rank bounds in the radial reduction are preserved at equality.

An exact-origin center has no polar angle. The graph therefore places no angular edges between it and other centers. Instead, since its distance to a center of radius s is exactly s, every other center must have radius at least 2. The refined enumerators enforce this directly: all other inner labels must be in bins with lower endpoint at least 2 (label 6 in the eight-bin grid and label 8 in the twelve-bin grid); the wall-pushed outer centers already have radius $b_0 > 2$.

For the positive first-bin type, $0 < x < 1/2$, the closed hull touches zero but does not represent the origin. When the other interval has upper endpoint at most 2, the cosine expression $c(x, y)$ is increasing in each variable on the relevant positive rectangle, since

$$\partial_x c = \frac{x^2 - y^2 + 4}{2x^2 y} \geq 0, \qquad \partial_y c = \frac{y^2 - x^2 + 4}{2xy^2} \geq 0.$$

Thus the value at the upper endpoints is a valid upper bound even as $x \downarrow 0$. If the other interval extends above 2, the verifier disables that angular edge. The exact-origin singleton is never subjected to this limiting-angle rule.

7 The local angular barrier

Assume the cyclic word is $(I, O, O)^5$. Let a_i be the five inner radii in cyclic order, with indices modulo five. Every outer radius is at most b_0.

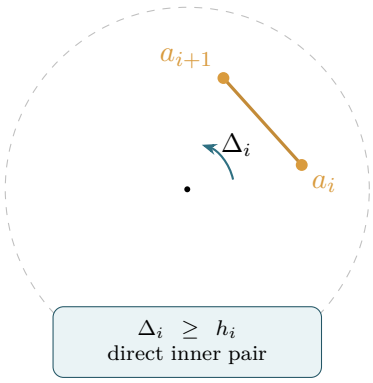
For each sector between consecutive inner-center directions, there are two independent routes that force it to use angular space. The direct chord between the inner centers gives one lower bound. The path through the two intervening outer centers gives another. Increasing the inner radii relaxes the direct inner-pair condition, but tightens the route through the outer centers; decreasing them reverses the roles. We take the larger bound so that either direction of radial change incurs an angular cost. The quantities below encode these two routes:

$$h_i = \Phi(a_i, a_{i+1}), \qquad k_i = \Phi(a_i, b_0) + \Phi(b_0, b_0) + \Phi(a_{i+1}, b_0).$$

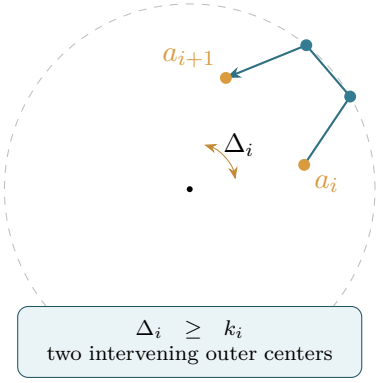
For $a_i \in [1.68, 1.72]$, the exact interval checks give $k_i > 1.175$. The four complementary inner-to-inner gaps therefore sum to more than $4.7 > \pi$, so the remaining gap is less than π. The direct inner pair and the path through the two intervening outer centers then give a lower bound $\max(h_i, k_i)$ on that gap. Summing the five gaps yields the necessary condition

$$\sum_{i=1}^{5} \max(h_i, k_i) \leq 2\pi. \tag{6}$$

When an outer center has radius below b_0, the inner–outer angle requirement is at least the value used in k_i, because $\Phi(a, s)$ decreases with s on the relevant interval. The outer–outer term obeys the analogous bound.



(a) The inner chord requires at least h_i.



(b) The three consecutive gaps sum to at least k_i.

Figure 6: Each inner-to-inner sector has two independent lower bounds. Combining them gives $\Delta_i \geq \max\{h_i, k_i\}$, which leads to the total angular budget in equation (6). The radii and angles are schematic.

Lemma 7.1 (Local angular rigidity)**.** *If $a_i \in [1.68, 1.72]$ for every i and the angular budget (6) holds, then $a_i = a_0$ for all i.*

Proof. At $a_i = a_0$, one has $h_i = k_i = 2\pi/5$. Write $\delta_i = a_i - a_0$ and $\sigma_i = \delta_i + \delta_{i+1}$. The direct route has first-order change $-c_0\sigma_i$, while the route through the two outer centers has first-order change $d_0\sigma_i$. Differentiation at the candidate gives

$$\partial_x \Phi(x, y)|_{(a_0,a_0)} = -c_0, \qquad \frac{d}{dx}\Phi(x, b_0)\Big|_{x=a_0} = d_0,$$

where

$$c_0 = \frac{\sin^2(\pi/5)}{\cos(\pi/5)} > \frac{21}{50}, \qquad d_0 = \cos(\pi/5) > \frac{1}{2} > c_0.$$

The second derivatives on $[1.68, 1.72]^2$ and on $[1.68, 1.72] \times \{b_0\}$ are bounded, with exact rational checks, by

$$|\Phi_{xx}(x, y)| < 0.3, \qquad |\Phi_{xy}(x, y)| < 0.8, \qquad |\Phi_{xx}(x, b_0)| < 5.$$

Taylor's theorem therefore gives

$$h_i \geq \frac{2\pi}{5} - c_0\sigma_i - \frac{5}{2}(\delta_i^2 + \delta_{i+1}^2),$$
$$k_i \geq \frac{2\pi}{5} + d_0\sigma_i - \frac{5}{2}(\delta_i^2 + \delta_{i+1}^2).$$

Since $d_0 > c_0 > 0$, $\max\{-c_0\sigma_i, d_0\sigma_i\} \geq c_0|\sigma_i|$: when one route becomes less restrictive, the other becomes more restrictive. The odd cycle length is essential. The linear map from the radial deviations δ_i to their consecutive sums σ_i is invertible; an even cycle would admit a nonzero alternating mode in its kernel. Indeed, the following identity holds:

$$2\delta_i = \sigma_i - \sigma_{i+1} + \sigma_{i+2} - \sigma_{i+3} + \sigma_{i+4}.$$

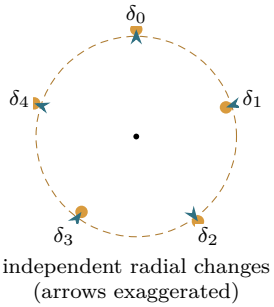
Taking absolute values and summing over all five indices gives

$$\sum_i |\sigma_i| \geq \frac{2}{5} \sum_i |\delta_i|.$$

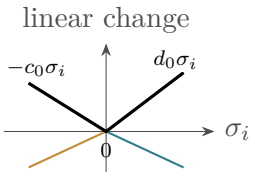
Moreover $|\delta_i| \leq 11/500$ throughout the local cube. If at least one δ_i is nonzero, the final inequality below is strict because $\frac{2}{5}c_0 - 5 \cdot \frac{11}{500} > \frac{29}{500}$:

$$\begin{aligned}
\sum_i \max(h_i, k_i) &\geq 2\pi + c_0 \sum_i |\sigma_i| - 5 \sum_i \delta_i^2 \\
&\geq 2\pi + \left(\frac{2}{5} c_0 - 5 \cdot \frac{11}{500} \right) \sum_i |\delta_i| \\
&> 2\pi + \frac{29}{500} \sum_i |\delta_i|.
\end{aligned}$$

This contradicts (6) unless every $\delta_i = 0$. □



(a) The five radii vary independently.



(b) Taking the larger linear term yields a positive penalty.

Figure 7: The local barrier has no hidden symmetry constraint: radial deviations are independent, while the maximum of the oppositely sloped first-order terms grows away from $\sigma_i = 0$. Quadratic errors are controlled by the exact interval bounds.

If every $a_i = a_0$ but $R < R_0$, each original outer radius is strictly below b_0. Its required inner–outer angle is then strictly larger than the candidate value, so the five-gap sum is strictly larger than 2π, again a contradiction. Thus no packing with $R < R_0$ exists in the final surviving class. Combining the explicit construction, the finite reduction, and the local angular rigidity lemma completes the proof of theorem 2.1.

8 Conclusion and reproducibility

The construction in section 3 proves $R_{15}^* \leq R_0$. The radial classification, exhaustive difference-constraint certificate, exact rational subdivision, and local angular barrier prove that every feasible packing has $R \geq R_0$, conditional on the finite certificate and geometric reductions stated above.

Replay files are in the repository's candidate bundle pinned to commit `02b6176`. Run it with Python 3.10 or newer, GNU C++17, and a POSIX shell:

```
cd problems/packing-15-equal-disks-in-a-circle/candidate_bundle
bash run_all.sh
```

All exclusion decisions use integers or rational fractions. The code's decimal output is diagnostic only. The replay is self-contained and records the branch counts and terminal reasons.

Subject to independent validation of the certificate and geometric reductions, this approach establishes the optimality of the fifteen-disk configuration without assuming a contact graph or symmetry.

A Certificate data and exact subdivision

The repository's candidate bundle pinned to commit `02b6176` contains the tables, branch rules, reports, and replay scripts. Stage zero writes `stage0_certificate.json.gz`, covering every excluded binary dihedral class and each radial type assignment not ruled out by rank constraints.

The independent `check_stage0_certificate.py` regenerates all 760 classes and checks assignment coverage and integer cycles. The replay also emits a 3.8 MB subdivision certificate with 1,266 roots. The independent `check_exact_certificate.py` verifies every bisection cover, pair-sum leaf, negative-cycle edge and weight, and local-region leaf using rational arithmetic; it checks the three $6I+9O$ assignments against `full_residuals.txt` and verifies all nine endpoint leaves. Stage counts and survivor identities are asserted during replay.

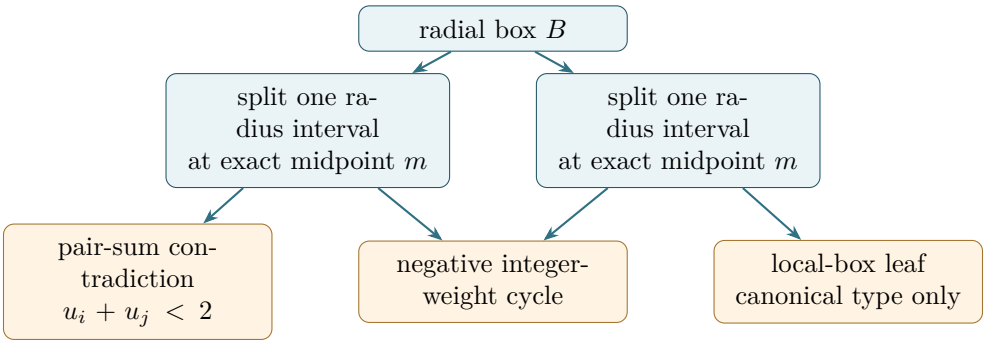


Figure 8: Each exact bisection covers its parent by two closed rational children, including the midpoint. A branch ends only at an exact contradiction or, for the canonical type, at a box passed to the separate local proof.

B Scope of the Lean formalization

The partial formalization is available in the repository's Lean project pinned to commit `69cfc63`. The project is hosted under the unequal-disk problem directory, but its `Fifteen` modules concern the equal-disk problem in this manuscript. They use the same candidate radii a_0, b_0 and the same local cube $[42/25, 43/25]^5$. The following distinctions are essential when interpreting the formalization status.

Manuscript component	Lean status at the pinned snapshot
Candidate construction (section 3)	Radius and contact-angle identities are proved. An explicit feasible `Packing 15` at radius R_0 remains to be constructed.
Radial reduction (section 4)	Wall replacement, the 5–8 inner-center count, and coarse radial-box membership are proved, together with canonical dihedral radius/angle normalization, complete Stage 0 assignment coverage, the sixth-radius bound, and refined half-open owner coverage.
Finite reduction (proposition 6.1)	All finite certificate replays are collected in one theorem. The geometric $760 \to 378$ reduction is proved. Packing-level soundness and coverage of Stage 3/4 and the refined subdivision remain unfinished; the four structural Stage 3 leaf families have geometric exclusion lemmas.
Local angular barrier (section 7)	Taylor error estimates, the odd five-cycle identity, and local rigidity are proved. The strict smaller-container exclusion is proved assuming the canonical cyclic pattern and the local-box hypotheses.
Optimality (theorem 2.1)	No unconditional Lean theorem of global optimality is present. It requires the packing-level finite reduction and the feasible candidate construction.

Replay and geometric soundness. The theorem `fifteenAllFiniteCertificates_replay`, in `CirclePacking/FifteenAllFiniteCertificates.lean`, combines the 760-class enumeration, coarse table and certificate checks, Stage 3 and Stage 4, the final rational subdivision, and the residual-pattern handoff. Its conclusion is a conjunction of finite check results and class counts. The separate theorem `fifteen_unit_packing_canonical_pattern_local_box_impossible`, in `CirclePacking/FifteenLocalFinalAssembly.lean`, excludes a packing with $R < R_0$ once its wall-replaced, angle-sorted profile has inner positions $0, 3, 6, 9, 12$ and those five radii lie in the local interval. The theorem `fifteen_unit_packing_wall_push_stage0_canonical_geometry`, in `CirclePacking/FifteenStageZeroCanonicalGeometry.lean`, now produces an exact canonical word and an ordered real polar profile from an arbitrary candidate-radius packing, preserving separation and coarse radius boxes and deriving the coarse angular difference constraints. Complete Stage 0 coverage and geometric exclusion are supplied by `fifteen_unit_packing_wall_push_stage0_survivor_geometry`, in `CirclePacking/FifteenStageZeroFiniteReduction.lean`. Every admissible positional label profile is proved to occur in the finite assignment enumeration, and its string encoding is proved to reconstruct the same fifteen labels. The replay then provides either a survivor record or a matching cycle witness; geometric cycle soundness rules out the latter. Thus an arbitrary feasible candidate-radius packing has a canonical profile among the 378 Stage 0 survivors. The theorem `fifteen_unit_packing_wall_push_stage3_geometric_handoff`, in `CirclePacking/FifteenStage3GeometricHandoff.lean`, attaches the refined owners, pair-sum bounds, and both rank counts to this same real profile. The owner proof covers the origin separately, assigns rational cutpoints to the interval on their right, and encloses every owner by its stored closed rational hull. Separate lemmas rule out the small-count, origin, rank, and mixed-bin leaves, including rank tests before all positions are assigned. The remaining Stage 3 step is to identify each checker state's labels and counters with its partial geometric assignment and prove cycle and stream soundness. Stage 4, the residual roots, and the six-inner special case still require their corresponding connections. Boolean replay alone does not supply these remaining implications.

Alternative intermediate proofs. The Lean development sometimes proves the same required bound by a different route. For the eight-inner-center upper bound, it uses a contact-cosine bound $c(x, y) \leq 2/3$ on $[1, 12/5]^2$ and $\arccos(2/3) > \pi/4$, together with the fact that at most one center has radius below 1. It therefore does not need the nine-disk optimum invoked in section 4. The 760 class counts are checked by exhaustive canonical-mask enumeration rather than by formalizing Burnside's formula. For local Taylor errors, Lean bounds the second derivative along a segment by $5(\delta_i^2 + \delta_{i+1}^2)$, giving the remainder in section 7 without individual Hessian bounds. The sixth-radius bound is also proved directly: for $0 < a, b < 2$, $a^2 + b^2 - ab \leq \max(a^2, b^2) < 4$, so any separated pair has contact angle greater than $\pi/3$. Six such centers would require more than a full turn. This retains the feasible boundary case at radius 2 and avoids assuming the six-disk optimum. The exact wall radius is proved to lie in the refined certificate interval $(35213569647/10^{10}, 35213569648/10^{10})$ by rational square comparisons.

Trust dependencies. Several finite replays use `native_decide` and carry native-evaluation axioms. Their trust dependencies must be distinguished from proofs using only ordinary kernel reduction. The abstract mask and angle transport lemmas and the local packing-exclusion theorem have been audited and depend only on Lean's standard logical axioms. The combined finite theorem can be built and audited from the Lean directory with:

```
lake build CirclePacking.FifteenAllFiniteCertificates
lake env lean CirclePacking/FifteenTrustAudit.lean
```

The audit prints the actual axiom dependencies; it is not a proof of the remaining geometric connection. The geometric normalization bridge has a separate targeted audit:

```
lake build CirclePacking.FifteenStageZeroCanonicalGeometry
lake env lean CirclePacking/FifteenStageZeroTransportAudit.lean
```

The packing-level normalization theorem inherits one existing native axiom for canonical coverage and seven for the coarse angle table. The normalized-profile cycle exclusion inherits one existing Stage 0 replay native axiom and the same seven table axioms; the mask and angle transport proofs introduce no new native-evaluation axiom. The Stage 0 coverage and geometric reduction have their own targeted audit:

```
lake build CirclePacking.FifteenStageZeroFiniteReduction
lake env lean CirclePacking/FifteenStageZeroReductionAudit.lean
```

Assignment coverage and the survivor count are derived from the existing Stage 0 replay rather than from an additional native-evaluation check. The full packing-level $760 \to 378$ theorem inherits nine existing native axioms: canonical coverage, the Stage 0 replay, and the seven coarse-table checks. Positional and transported population-count lemmas use only the standard logical axioms. The refined owner and structural-leaf interface has a further targeted audit:

```
lake build CirclePacking.FifteenStage3GeometricHandoff
lake env lean CirclePacking/FifteenStage3GeometryAudit.lean
```

The sixth-radius bound, refined wall enclosure, owner coverage, partial-rank and structural exclusion lemmas, and packing-level structural conditions depend only on the standard logical axioms. The combined Stage 3 input handoff inherits the same nine Stage 0 native axioms as the $760 \to 378$ reduction; these new proofs introduce no native-evaluation axiom.

References

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